IC-252 Lab Assignment 5

Due for Moodle submission on 16th April

Question-1:

A data file named ‘random_num’ has been provided containing an array of 10000 random real numbers.

1) Write a Python code to:

a) Find the mean and standard deviation.

b) Find the distribution or histogram (take bin size 0.02) of the given data and mention the most probable bin.

c) Identify the smaller group of outlier data and filter it out.

d) For the remaining data, find the mean and variance.

e) Find the distribution of this remaining data.

Question-2:

Covid-19 proved to be a deadly pandemic which caused millions of deaths and destroyed the economy of various nations. Lockdowns were imposed to contain the spread of the virus. Mobility was significantly reduced during this period.

The Community Mobility Reports provided by google show movement trends by region, across different categories of places. These reports are created with aggregated, anonymized sets of data from users who have turned on the Location History setting, which is off by default. The data shows how visitors to (or time spent in) categorized places change compared to our baseline days. A baseline day represents a *normal* value for that day of the week. The baseline day is the median value from the 5-week period Jan 3 – Feb 6, 2020.For each region-category, the baseline isn’t a single value—it’s 7 individual values. The same number of visitors on 2 different days of the week, result in different percentage changes.

You are given a dataset which contains the mobility data of India in 2021. You are expected to do the following for Mumbai City:

A. Plot the monthly graph (Month Name on X-axis) for each category of mobility (grocery,retail,transit,parks,residential,workplaces). What do you infer from these graphs?
B. Calculate the variance for each of the 6 mobilities for 2 time periods :
    - During the lockdown period (consider only the month of April and May (till 20/05/2021)).
    - For the entire year (365 days).

    Compare these variances for each category. What do you infer ?

C. Let’s assume that the data given by Google is probabilistic in nature and each mobility has a certain probability associated with it. So on a particular day for a region we compute the expected mobility using the formula :

$$E(Mobility) = \sum P(Mobility_i) \times Mobility_i$$

Consider the following distribution during the lockdown period (1/04/2021 to 20/05/2021):

Grocery / Pharma	Retail	Transport	Parks	Residential	Workplace
p=0.2	p=0.2	p=0.05	p=0.02	p=0.5	p=0.03

Compute the expected mobility for each day during the lockdown period and compare these values with the original mobilities for each category during the lockdown period. What do you infer ?

Question 3:

Consider the following data for two stocks A and B:

Scenario	Probability	Asset A Return (%)	Asset B Return (%)
I	0.4	-10	5
II	0.2	10	20
III	0.4	20	10

1) Theoretically compute the expected return, variance, and standard deviation for each stock.
2) For each stock randomly sample N return values following the above given probability distribution. For simulation take N = 100,200,500, and 1000. For each value of N, from the sampled return values compute mean, variance and standard deviation and compare them with the theoretical results for each stock.
3) Suppose that the investor makes a portfolio which invests w1 proportion of the wealth in stock A and the remaining (lets say w2) in stock B. For (w1=0.4, w2=0.6), (w1=0.5, w2=0.5) and (w1=0.8, w2=0.2) theoretically compute the expected portfolio return and risk. Based on the expected risk and return value which (w1, w2) combination you will prefer for investment.
4) For your preferred (w1, w2) combination compute the portfolio return and risk from the sampled return data in step 2 for N=1000. Compare the theoretical and results from sampled data.